A cross-cultural analysis of toolmaking skill acquisition in non-industrial societies

Cheng Liu

2025-12-28

Contents

1 Introduction

1.1 Toolmaking skill acquisition: A cross-cultural perspective

Toolmaking is critical to human experience because it creates a biocultural feedback loop that continuously shapes both our cognition and physical anatomy (Birch, 2021; Malafouris, 2012, 2021; Stout, 2021). Neuroimaging studies (Stout & Chaminade, 2007; Stout & Hecht, 2024; Stout & Khreisheh, 2015) have demonstrated how stone tool technologies influenced our brain development and cognitive capabilities, while biomechanical research (Dunmore et al., 2023; Karakostis et al., 2021; Kivell et al., 2023; Marzke, 2013) reveals how tool production and use drove significant changes in human hand morphology over evolutionary time. This reciprocal relationship between toolmaking and biological adaptation represents a uniquely human pattern of niche construction that has fundamentally shaped our species' evolution.

Yet the skill of toolmaking is not born with us—naive individuals need to acquire a given skill either through observing and learning from others (social learning) or repeated trials and errors (individual learning). Very often skill attainment, particularly this type of motor skill involving extensive hand-eye coordination, requires a combination of these two learning strategies. Although this behavior is of great significance in human evolutionary studies (Buckley, 2023; Derex & Morgan, 2023), the mass machinery manufacturing process of tools introduced by industrialization weakens the necessity of handicraft practices and their inter-generational transmission in our contemporary daily life, often accompanied by the widespread formal schooling (Becker & Woessmann, 2024) following a European model that masked varying learning contexts in human societies. Consequently, ethnographic narratives depicting the subsistence and economy of non-industrial societies, mostly written during the nineteenth and twentieth centuries, provide us a precious opportunity to investigate this topic, among which hunter-gatherer societies, featuring complex and diverse modes of social network (Bird et al., 2019) and political organization (Wengrow & Graeber, 2015), represent the most well-studied and well-published sample.

Embedded within the century-old anthropological research tradition revolving around childhood, the learning dynamics of hunter-gatherers have garnered considerable attention in the past two decades due to the burgeoning literature of cultural evolutionary theories (Hewlett et al., 2024). Garfield et al. (Garfield et al., 2016) conducted the first systematic investigation on hunter-gatherer social learning, involving 982 ethnographic texts from 23 different communities. In this research, they characterized the commonalities of learning behaviors across diverse domains regarding their transmission mode, forms of teaching, and learners' age. Within the acquisition of subsistence skills, which dominated early and middle childhood, they concluded that oblique transmission becomes more important when compared with vertical transmission with the increasing age of learners. Subsequently, Lew-Levy and her colleagues narrowed down the learner's identities to children and thoroughly examined their learning patterns of subsistence skill, of which toolmaking skill is a subset, depicting a more detailed picture regarding the dominant learning processes and skill types across each developmental period (Lew-Levy et al., 2017; Lew-Levy, Milks, et al., 2020).

Drawing upon both the cross-cultural analysis of existing ethnographic texts and new systematic field data collected from certain foraging communities, this children-centered research line of inquiry (Lancy, 2017) remained the most fruitful one in the past few years, emphasizing the evolutionary significance of object/toy as a medium of learning (Lew-Levy et al., 2022; Riede et al., 2023) and the often ignored strategy of peer-learning (Lew-Levy, Kissler, et al., 2020; Lew-Levy et al., 2023; Lew-Levy & Amir, 2024). In addition, other researchers conducted either literature surveys (Bira & Hewlett, 2023) or in-depth systematic field observations (Schniter et al., 2023) on the knowledge transmission patterns of societies featuring other forms of subsistence strategies. For example, Bira and Hewlett (2023) compared the cultural learning patterns between hunter-gatherers and pastoralist children and found that peer-learning was less likely to be documented in the latter compared with the former context. Based on long-term field data of Tsimane forager-farmers, Schniter et al. (2023) identified the importance of older same-sex relatives and multigenerational pedagogy in the successful reproduction of crucial skills like handicrafts.

Some pressing issues still remain unsolved despite previous intellectual efforts on the topic of toolmaking skill acquisition. First and foremost, the extremely low contribution of individual learning in subsistence skills, particularly toolmaking, which according to Garfield et al. (2016) only occupies 3.7% of all cases. There could be several reasons behind it. An obvious one is that ethnographers may systematically care more about communal activities and it is challenging to follow and observe individuals, particularly when they conduct foraging activities outside the settlement (e.g., Hulstaert, 1938: 476). Another potential source of bias could be the coding process, which I aim to improve in this project. I have developed a coding scheme that provides a more inclusive evaluation of the learning processes. Second, the unbalanced representation of hunter-gatherer societies versus all other subsistence types within non-industrial societies masks our understanding because toolmaking (by hand) is not unique to hunter-gatherers.

1.2 Study aims

To understand the diversity and regularity in the process of human toolmaking skill acquisition, this project analyzed 170 non-industrial societies displaying varying subsistence strategies around the world using the eHRAF World Cultures database, with a particular focus on developmental contexts and transmission biases involved in the learning processes of toolmaking skills. This study is exploratory and descriptive in nature, focusing on systematically summarizing and presenting the available data without attempting to infer causal relationships or test predictive models. Descriptive studies are essential for establishing a foundational understanding of complex datasets, identifying patterns, and guiding future research directions by highlighting key trends and data limitations (Gerring, 2012). This study investigates the processes of toolmaking skill acquisition in non-industrial societies, focusing on fundamental aspects of learning: what, who, where, when, how, what, and why. The "what" focuses on the content of transmission, such as pottery-making, woodworking, or other craft activities. For the "who" aspect, the author examined the transmission modes and concrete social relationships between the learner and the model. The "where" question addresses the spatial context of learning, such as whether it occurs indoors or outdoors. The "when" explores the ages of both the learner and the model. For the "how," the author investigated the degree of social interaction involved in the learning process. Lastly, the "why" examines the motivations behind a learner's choice of model—whether it is based on the model's recognized expertise, convenience, or familial ties.

In addition to addressing the six fundamental research questions, this project also undertakes a missing data pattern analysis to evaluate the completeness and reliability of the dataset. Furthermore, a qualitative analysis of the ethnographic texts was conducted to provide contextual depth and enrich the interpretation of the database. Together, these approaches offer a more comprehensive understanding of the data, enhancing both the analytical rigor and the interpretive framework of this study.

2 Methods and materials

2.1 Searching strategy

This study analyzed a global sample composed of 170 non-industrial societies indexed in the eHRAF World Cultures database, which include 45 hunter-gatherers, 36 horticulturalists, 29 other subsistence combined, 28 intensive agriculturists, 14 primarily hunter-gatherers, 13 agro-pastoralists, and 5 pastoralists cultures (**Figure** 1). Societies labeled as commercial economy are pre-excluded here since they are often associated with industrialization to some degree. Because the eHRAF database is organized using the Outline of Cultural Materials (OCM) coding scheme, the searching strategy focuses exclusively on one OCM code that is most relevant to the research question, namely OCM 868 "Transmission of Skills". To identify relevant paragraphs within this OCM code, we conducted several individual searches in the eHRAF database. Each search used a single keyword. The keywords used in these separate searches were: tool, make, made, making, produce, producing, manufacture, manufacturing, create, creating, construct, constructing, assemble, assembling, fabricate, fabricating, shape, shaping, forge, forging, form, and forming. This multi-search process yielded a total of 3060 paragraphs. Because we conducted separate searches, the same paragraph was often returned multiple times if it contained more than one of our keywords (e.g., a paragraph containing both "make" and "tool"). This method resulted in an estimated 30–40% of the initial results being duplicates. We removed all duplicate paragraphs before beginning the screening process.

Focusing on the toolmaking skill acquisition in non-industrial societies, this project adopts the following exclusion criteria when screening the paragraphs yielded in the initial search, including archaeological and historical literature sources, cases of purely tool use, instances where local people taught the ethnographer how to make tools, instances where it was clearly stated a new form of government-initiated school to save a certain craft, instances where individuals are taught how to make Christianity-related artifacts such altar, and instances where indigenous people mentioned that they learn a specific craft from museum pieces or photographs. These instances, occurring in very low frequency, represent either non-traditional learning contexts or documentation methods that do not capture the "natural" processes of skill transmission within these communities as a result of colonization and/or market integration. However, these cases may appear as contextual information in the Supplement to provide more details of the technologies themselves. In the end, the eligible paragraphs and their associated contextual paragraphs were compiled as a geographically organized Supplement of this article to ensure full transparency of the source data.

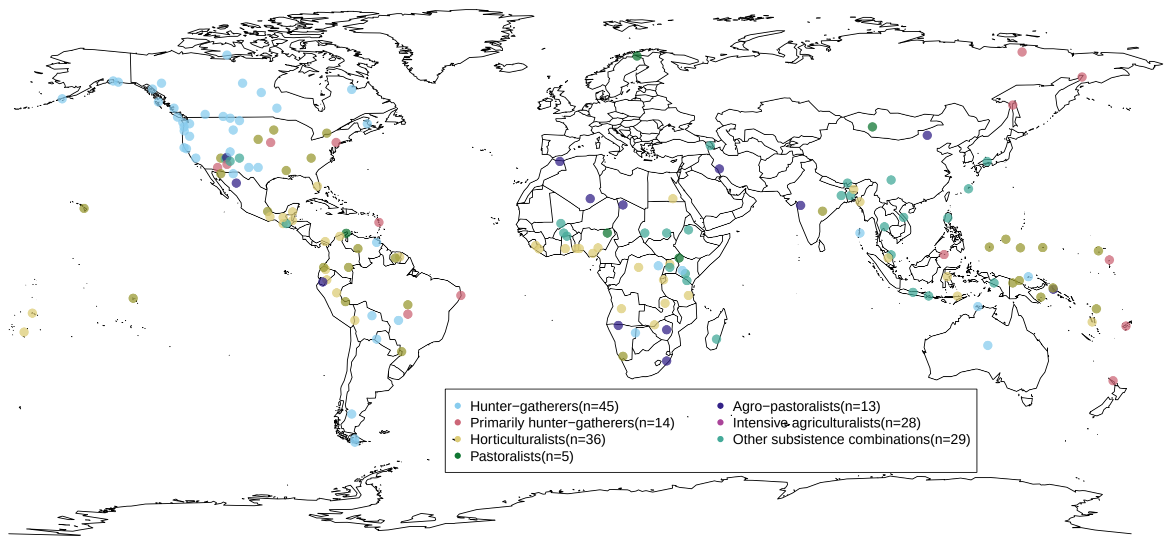

Figure 1: The spatial distribution of non-industrial societies covered in this study (n=170). It should be noted that these coordinate data represent language 'centroids' from the open-access language documentation site glottolog.org instead of the real geographical locations of given communities. These language centroids essentially represent the center point of the area where a relevant language is spoken.

2.2 Coding scheme

In the coding process of ethnographic texts, the acquisition of a specific toolmaking skill (e.g., clothes making) documented in one culture is regarded as an instance (i.e., a single row in the spreadsheet). When multiple sources by either the same or different authors all mentioned relevant information regarding how certain technology is transmitted in a society, the coding scheme treated all of them as "true" information, assuming that it is possible for a technology to have multiple channels of transmission unless explicitly stated otherwise. For example, source

A claims that pottery making in society X is transmitted between parents and offspring, while source B claims that pottery making in the same society is transmitted between grandparents and grandchildren. In this case, the transmission mode of this observation/row should be coded as “mixed” instead of “vertical” or “oblique”. The only exception is the age group of learner, for which the lowest age group is adopted (see the full coding scheme below for more details).

To ensure the reliability of coding results, the whole Supplement was coded twice by the author until all discrepancies were resolved. The author employed a dual-system recording approach with some key variables such as technology, location, age of learners and models, learning process, etc. Specifically, the author first documented the descriptions provided by ethnographers and then categorized the information in a separate column for quantitative analysis. It should be noted that “NA” is used in all variables when there is not enough information to make the relevant inference, which is different from the absence of the trait of interest. The following is a complete description of the coding scheme used in this project. To illustrate how the coding scheme was applied in practice, here I also provide several examples related to the “how” and “why” questions, which are arguably the more subjective aspects of the coding process.

1. **Area**: This variable follows eHRAF’s classification system.
2. **Society**: This variable follows eHRAF’s classification system.
3. **OWC Code**: The Outline of World Cultures (OWC) is a classification system of the cultures of the world, which is used by eHRAF to organize its databases. It is used here as a unique identifier to merge the geographic coordinate data from D-PLACE for the purpose of mapping.
4. **Subsistence**: This variable follows eHRAF’s classification system, including hunter-gatherer, primarily hunter-gatherer, horticulturalist, pastoralist, agro-pastoralist, intensive agriculturalist, and other subsistence combinations. The category of commercial economy is excluded. The definitions of these terms are available in the official eHRAF documentation here (https://ehrafworldcultures.yale.edu/help/LearnMore/subsistence.html).
5. **Year of observation**: The years when the given society was observed by the ethnographers. This information is readily available under the name “field date” in the meta-information of eHRAF World Cultures.

6. **Location**: The location where the learning event happens according to the ethnographer's description.

7. **Location (type)**: 1) *indoor:* learning happens inside a building; 2) *outdoor:* learning happens outside the buildings.

8. **Age of learner**: The learner's age according to the ethnographer's description.

9. **Age group of learner**: 1) *Infancy and Early Childhood* (0-5 years old), 2) *Middle Childhood* (6-12 years old), 3) *Adolescence* (13-16 years old), 4) *Adulthood* (>16 years old). For analytical purposes, this category does not concern the cultural differences in the perception of developmental differences. When an age range is provided, the minimal age will be used in the classification of age group.

10. **Number of learners**: The number of learners involved, which can either 1) *single* or 2) *multiple*. When sources provided conflicting evidence (e.g., one source mentioned a single learner and another mentioned multiple), the code was defaulted to "multiple."

11. **Age of model**: The model's age according to the ethnographer's description.

12. **Age group of model**: 1) *Infancy and Early Childhood* (0-5 years old), 2) *Middle Childhood* (6-12 years old), 3) *Adolescence* (13-16 years old), 4) *Adulthood* (>16 years old).

13. **Number of models**: The number of models involved, which can either 1) *single* or 2) *multiple*. See **Number of learners** when sources provided conflicting evidence.

14. **Transmission modes**: 1) *horizontal transmission*: social learning from individuals of the same generation, age group, or cohort, roughly within 4–5 years of age; 2) *vertical transmission*: children learning from their parents; 3) *oblique transmission*: social learning between individuals of distinct generations or age groups typically from an older generation to a younger generation. It can happen within or beyond extended kin groups, such as learning from grandparents, uncles, or other adults in the local population; 4) *mixed*: different transmission modes occurred together in the learning process of a given technology in a single source or multiple sources.

15. **Social relationship**: All types of relationships between the learner and the model according to the ethnographer's description. For example, mother-son/father-daughter/etc.

16. **Transmission bias**: This category follows the framework developed by Kendal et al. (2018). 1) *kin-based bias*: learning a cultural trait from people who are your kin members. For example, among Ganda agriculturalists, "Specialists, such as smiths and potters, taught their sons their crafts (Mair, 1934: 65)." 2) *success-based bias*: learning a cultural trait from people because they are successful in this technology. For example, for Berbers of Morocco, "a girl who is not expert invites a number of skilled weavers, who go to her house, eat, and instruct her (Coon, 1931: 77)." 3) *prestige-based bias*: learning a cultural trait from people because of their high social status. For example, in an ethnographic narrative of Mescalero Apache people, it was stated that "He was a chief when I came back from being a prisoner of war... Natsili was really a good leader in all things. He showed us how to hunt, to ride horses, to make saddles, bows and arrows, ropes, quirts, head bonnets, shields, and many other things (Opler, 1969: 59, 61)." 4) *familiarity-based bias*: learning from the people they interact with most frequently, simply because of that repeated exposure, particularly playmates. For example, among Tallensi people, "...whereas the young men and women plait sporadically a few things for themselves, their sweetheart, or a child, children do so continuously and absorbedly. From July to September one sees them sitting about or strolling about in small groups, plaiting reed decorations (Fortes, 1938: 60-61)." 5) *mixed*: different transmission biases occurred together in the learning process of a given technology in a single source or multiple sources. For example, kin-based and success-based transmission biases co-exist in an instance of pottery skill transmission among Nuba people. "The factory occasionally turns into a school for potters. For (grown-up) women can come and watch the experts at work; they can help with the simpler tasks and will be instructed—free of charge—in the more difficult ones, till they have thoroughly mastered the technique and are ready to open a workshop of their own. Otherwise the craft is handed on from mother to daughter (Nadel, 1947: 71-72)." It should be noted that successful or prestigious toolmakers most likely also have close kin members, including offspring. When both identities (e.g., as a kin member and as an expert) of the model are explicitly mentioned, these instances are coded as transmission bias. Lastly, conformist bias, namely learning a specific type of knowledge from other people because it is displayed by the majority of the local population, is absent in our example.

17. **Learning process**: The learning process according to the ethnographer's description, such as "teach", "imitate", "observe", etc.

18. **Learning process type**: 1) *individual learning*: Individual exhibits repeated attempts to learn a skill or develop new skills or knowledge on his or her own, including trial and error and individual practice. For example, according to a Tiv mother: "I've given her some clay, and she'll make a very small pot. But she's got a lot of growing to do before she can really learn anything. (Bohannan, 1958: 381)." 2) *observational learning*: learner directly observes the actions of models and attempts to replicate the observed actions, and no direct interaction between the model and learner is observed or mentioned. For example, among Yoruba horticulturalists, "By watching his father he learns and by the time he reaches the age of sixteen he will have enough knowledge to perform most tasks and needs only a little practice (Lloyd, 1953: 33)." 3) *interactive learning*: this category features the verbal, gestural and bodily interactions between models and learners, be it formal ("teaching") or informal ("group play"). For example, among Bhil agro-pastoralists, "Children are taught work techniques by being instructed to copy their elders, by verbal instructions, or in easy works by asking them to help, as in rope-making (Naik, 1956: 119)." It should be noted that these three categories are not mutually exclusive. On the contrary, these three categories are inclusive of each other, where interactive learning is built on both observation and individual practice, and observational learning involves individual practice.

19. **Time spent**: The time spent on learning according to the ethnographer's description.

20. **Technology**: The subject of learning according to the ethnographer's description.

21. **Technology type**: The types of technology were determined *post hoc* from collected ethnographic texts based on their frequency, including agriculture equipment making, bag making, basket making, boat making, clothes making, fishing devices making, general toolmaking, hammock making horse harnesses making, mat making, metal working, miscellaneous, musical instrument making, net making, ornament making, pottery making, raw material processing, ritual item making, rope making, stone working, toy making, trap making, vehicles making, vessel making, weapon making, weaving, woodworking. These categories are not mutually exclusive, and some categories are singled out either because of cultural significance or the lack of relevant information. For example, weaving is a commonly documented skill that can be used in the manufacture of clothes, bags, baskets, hammocks, and mats, but it has its own category because no other contextual information is available to differentiate between them. Miscellaneous is used when the frequency of a

specific toolmaking skill is lower than three. General toolmaking is an umbrella term for phrasing like "utensils" or "tools" in the original text.

22. **Raw material**: The raw materials were also determined *post hoc* from collected ethnographic texts based on their frequency, including stone, clay, plant fiber, wood, bones, leather, composite, etc.

23. **Note**: This records interesting information that is not systematically documented in other variables, such as the difficulty of a skill as perceived by local people and ethnographers, access to raw material, and the restriction on the social status involved in the learning such as gender, wealth, religious status, etc.

24. **Reference**: This provides the relevant references, which can be linked to the relevant paragraph in the Supplement.

2.3 Taphonomic Control: Clay-Specific Analysis

In addition to the broader analysis, this study also investigates how differential preservation of raw materials might affect our ability to infer toolmaking acquisition patterns from material culture remains. Following the primary analysis, I selected clay materials—one of the most common archaeological remains and the only inorganic material with more than 30 documented instances—and visualized these data alongside the main results in terms of the "who", "when", "how", and "why" questions. The standalone "where" analysis is excluded here for clay materials because of its high missing value. This approach helps address potential preservation bias while also mitigating overrepresentation issues present in the general dataset, as certain societies receive disproportionate ethnographic attention regarding craftsmanship and childhood learning. For example, Lancy's (1975, 1996) work on Kpelle horticulturalists have documented more than 15 different toolmaking skills in the dataset, whereas most societies only feature one or two skills. When we narrow down to a single skill category, each society is represented by only a single entry as stated in the coding scheme below. Therefore, this focused analysis of clay materials provides an important methodological control against these sampling biases.

2.4 Statistical analysis and visualization

To assess the patterns of missing data, the author conducted a comprehensive missing data analysis using a combination of visualization techniques and statistical testing. A heatmap and an UpSet plot were generated with the R package **naniar** (Tierney & Cook, 2020) to visually explore the distribution and intersections of missing values across variables. The heatmap highlighted the overall distribution of missing data, while the UpSet plot provided insights into how missingness co-occurs among specific variables. Additionally, Little's MCAR test was performed to statistically evaluate whether the missing data were missing completely at random (MCAR). To ensure transparency and reproducibility (Marwick, 2017), the dataset and the full analysis code have been uploaded to a dedicated GitHub repository (https://github.com/Raylc/HRAF-2022).

3 Results

3.1 Descriptive results

3.1.1 Content of transmission: What skills are learned?

Figure 2A reveals significant variation in the representation of different craft activities in non-industrial societies. Clothes-making emerges as the most frequently recorded skill (n = 64), followed by basket-making (n = 49), weapon-making (n = 44), and pottery-making (n = 38). Weaving (n = 28) and woodworking (n = 26) are also prominently represented. In contrast, certain specialized skills, such as vehicle-making and agricultural equipment-making, appear infrequently (n = 3 each). This distribution suggests that daily utilitarian crafts, particularly those related to clothing and storage, are more commonly documented, possibly reflecting their widespread necessity and frequent practice. In contrast, more specialized or context-specific crafts, such as ritual item-making (n = 6) and horse harness-making (n = 4), are less commonly reported. The variation in frequency may reflect the cultural importance of certain skills and/or the observational bias focusing on more pervasive or visible activities within these societies.

If we reorganize the classification of toolmaking skills based on the raw materials involved, the results reveal a significant disparity in their distribution (**Figure** 2B). The majority of entries fall under the "NA" category (n = 182), indicating a substantial portion of missing data for this variable. Among the recorded materials, plant fiber (n = 157) and wood (n = 63) are the most frequently documented, suggesting their prevalent use or better preservation in the archaeological record. In contrast, materials such as clay (n = 39), metal (n = 17), leather (n = 16), stone (n = 7), composite (n = 4), and bone (n = 2) appear far less frequently.

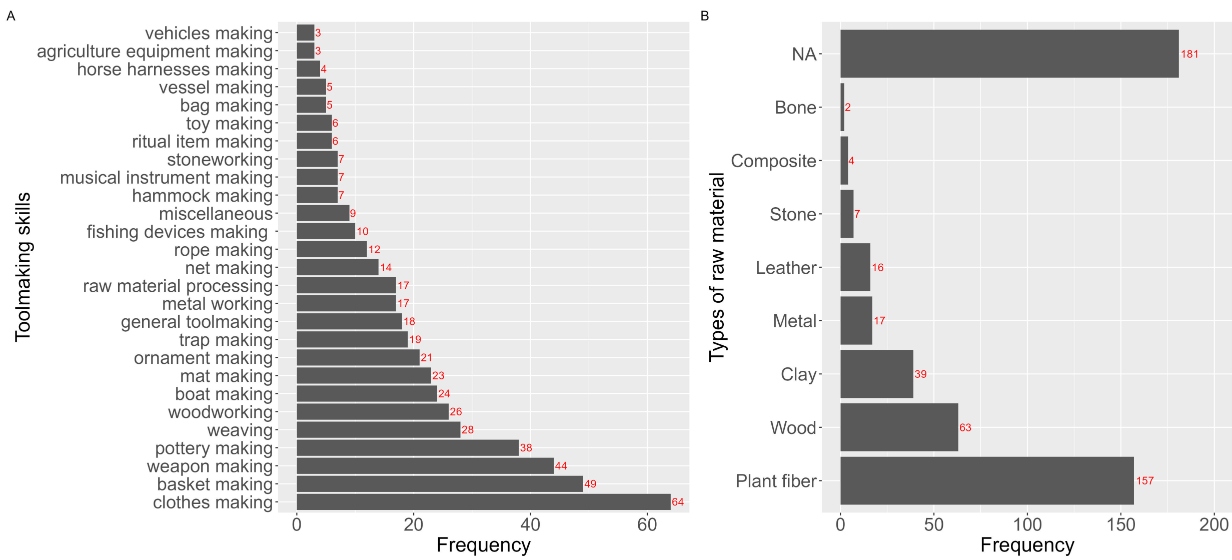

Figure 2: The frequency of toolmaking skills (A) as well as main raw materials involved (B) as documented in ethnographic texts.

3.1.2 Agents of transmission: Who transmits skills?

The results (**Figure** 3A) indicate that vertical transmission, where skills are passed from parent to child, is the most frequently documented mode (n = 162). This is followed by vertical/oblique transmission (n = 112), where learning occurs both from parents and other adults within the community. Purely oblique transmission—learning from non-parental adults—accounts for 59 instances. Horizontal transmission, involving learning from peers, appears less frequently (n = 10), and mixed modes combining horizontal and oblique transmission are even rarer (e.g., horizontal/oblique: n = 6; oblique/horizontal: n = 1). Additionally, some cases report a combination of all modes (n = 11). Notably, 118 instances lack specific information on transmission mode, highlighting gaps in ethnographic reporting. As for the clay-specific analysis shown in **Figure** 3B, the distribution pattern largely mirrors the overall trend with the exception of missing instances of the "All" category (all three modes present), where the vertical transmission remains predominant (n = 14).

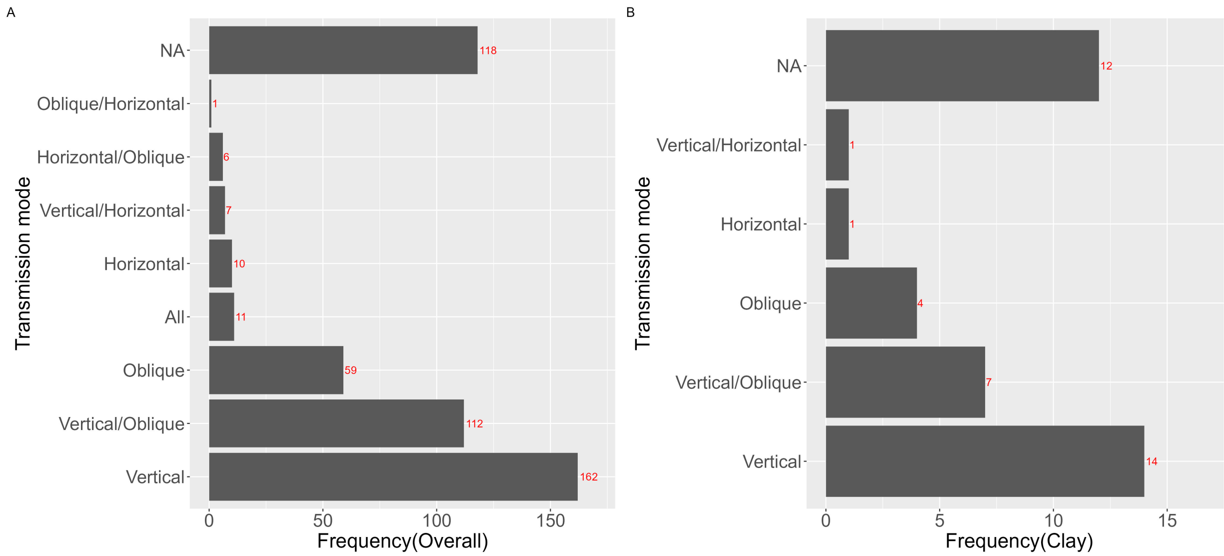

Figure 3: Transmission modes of toolmaking skills documented in ethnographic texts.

3.1.3 Spatial context: Where does learning occur?

The majority of ethnographic cases (n = 421) lacked explicit information regarding the learning locations of toolmaking skills (**Figure** 4A). Among the cases with documented locations, learning occurred most frequently indoors (n = 38), followed by outdoor settings (n = 25), with only two instances involving mixed indoor-outdoor contexts. **Figure** 4B further illustrates the relationship between learning environments and specific tool-making activities using a treemap. Indoor learning environments are predominantly associated with activities such as clothes making (n = 6), basket making (n = 5), and metal working (n = 5). In contrast, outdoor learning environments are primarily linked to boat making (n = 4), trap making (n = 3), and net making (n = 2). Only two activities, metal working, and stoneworking, were reported in both spatial contexts.

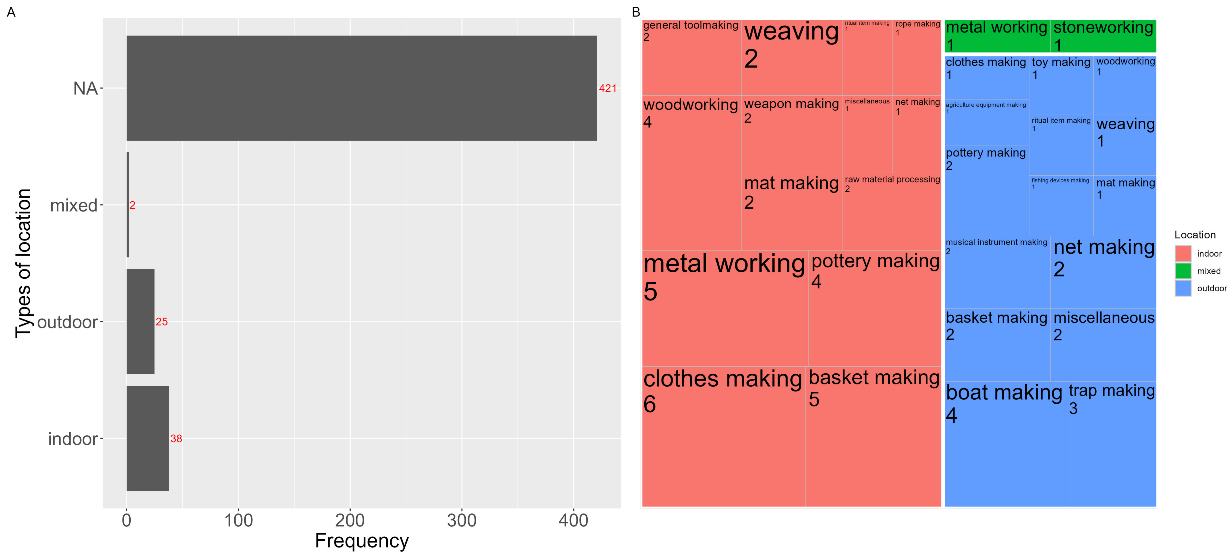

Figure 4: Location of toolmaking skill acquisition documented in ethnographic texts. A displays the frequency of learning locations. B is a treemap demonstrating the connections between craft categories and location locations.

3.1.4 Developmental timing: When are skills acquired?

The overall analysis of toolmaking skill learners' developmental contexts (**Figure** 5A) revealed that most learning instances (n=274) did not have specific age-related information documented. Among the cases where age was specified, middle childhood (ages 6-12) showed the highest frequency of learning episodes (n=128), followed by adolescence (ages 13-16) with 47 instances. Infancy and early childhood (ages 0-5) accounted for 34 documented cases, while adulthood showed the lowest frequency with only 3 recorded instances. The clay-specific analysis in **Figure** 5B follows a similar pattern to the overall results, with middle childhood again representing the most frequent learning period (n=7), though the proportional distribution differs slightly with relatively more cases documented in infancy and early childhood (n=6) compared to adolescence (n=2).

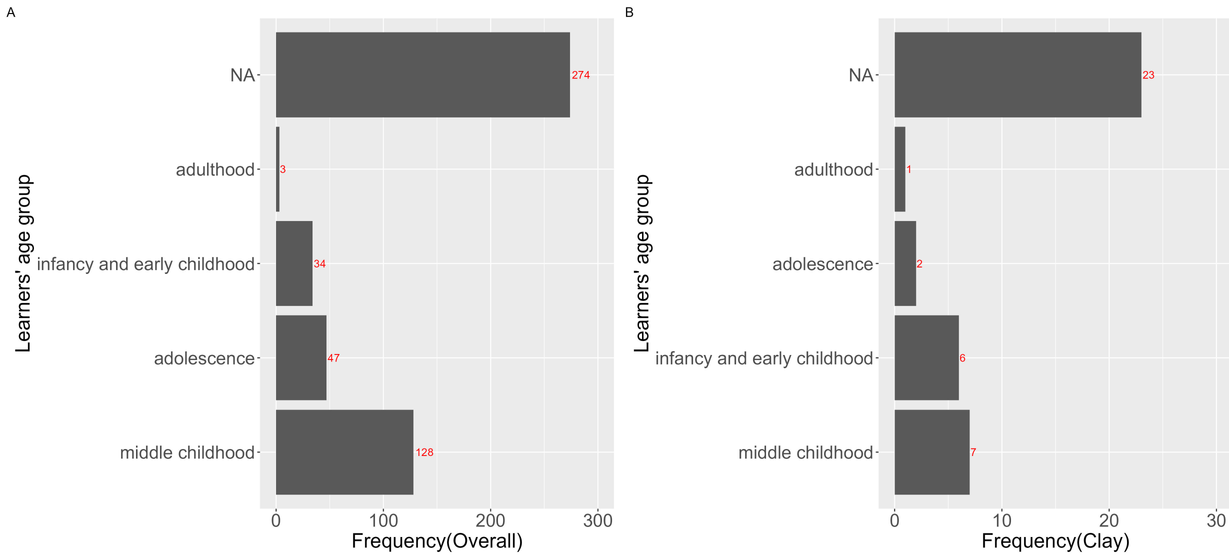

Figure 5: Developmental contexts of toolmaking skill acquisition documented in ethnographic texts.

3.1.5 Learning processes: How are skills learned?

Figure 6A presents the distribution of learning processes for toolmaking skills as follows. Interactive learning, which involves direct engagement between teacher and learner, was the most common method documented with 304 instances. Observational learning, where individuals learn by watching others without direct interaction, occurred in 62 cases. Individual learning, characterized by self-directed experimentation and practice, was recorded in 16 instances. In 104 cases, the specific learning process was not identified. The clay-specific analysis (**Figure** 6B) shows a proportional distribution that closely mirrors the overall pattern, with interactive learning strongly dominating (n=23) compared to observational learning (n=6) and individual learning (n=3).

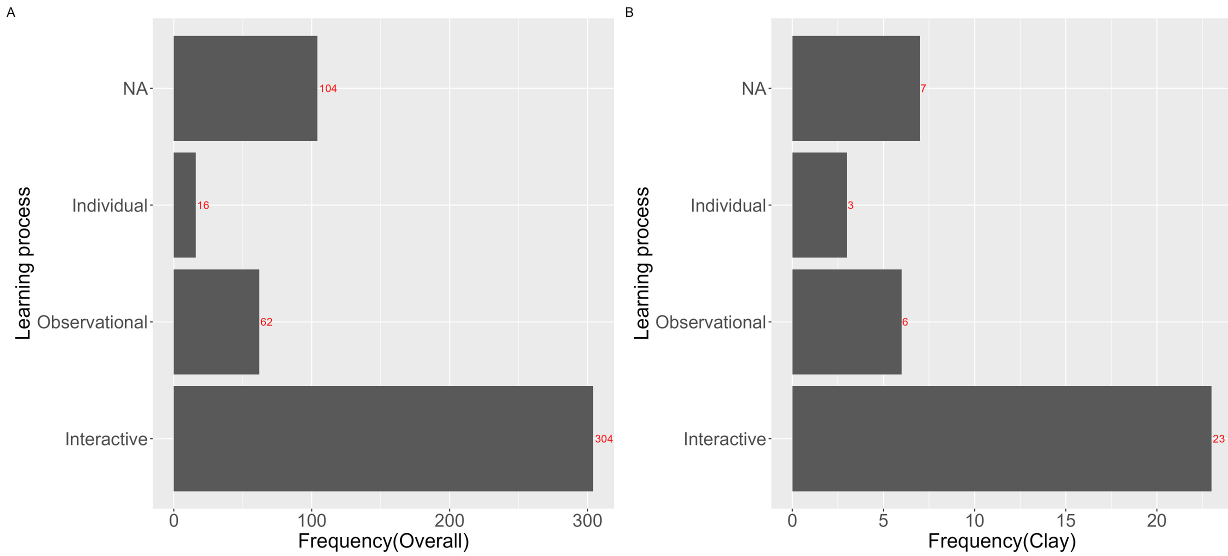

Figure 6: Learning processes of toolmaking skill acquisition documented in ethnographic texts.

3.1.6 Transmission biases: Why are models chosen?

The distribution of transmission bias (**Figure** 7A) agrees with the result of transmission mode analysis as illustrated in the who section above. Kin-based transmission emerged as the predominant pattern with 288 documented cases, suggesting that tool-making skills are most often passed down within family units. Mixed transmission strategies, which combine different learning approaches, were observed in 33 cases, while success-based transmission occurred in 28 instances. Prestige-based transmission, where learners chose models based on their social status, was documented in 8 cases, and familiarity-based transmission appeared in only 3 instances. A substantial number of cases (n=126) did not specify the transmission bias (NA). Again, this distribution shows that kinship plays a central role in the cultural transmission of tool-making skills, while other transmission biases occur less frequently. In the clay-specific analysis shown in **Figure** 7B, the dominance of kin-based transmission is even more pronounced (n=26), representing a higher proportion of the total clay cases compared to the overall dataset. Meanwhile, prestige-based transmission is entirely absent in the clay subset, while mixed (n=2) and success-based (n=1) transmission appear at lower frequencies, potentially because the subset sample size is too small.

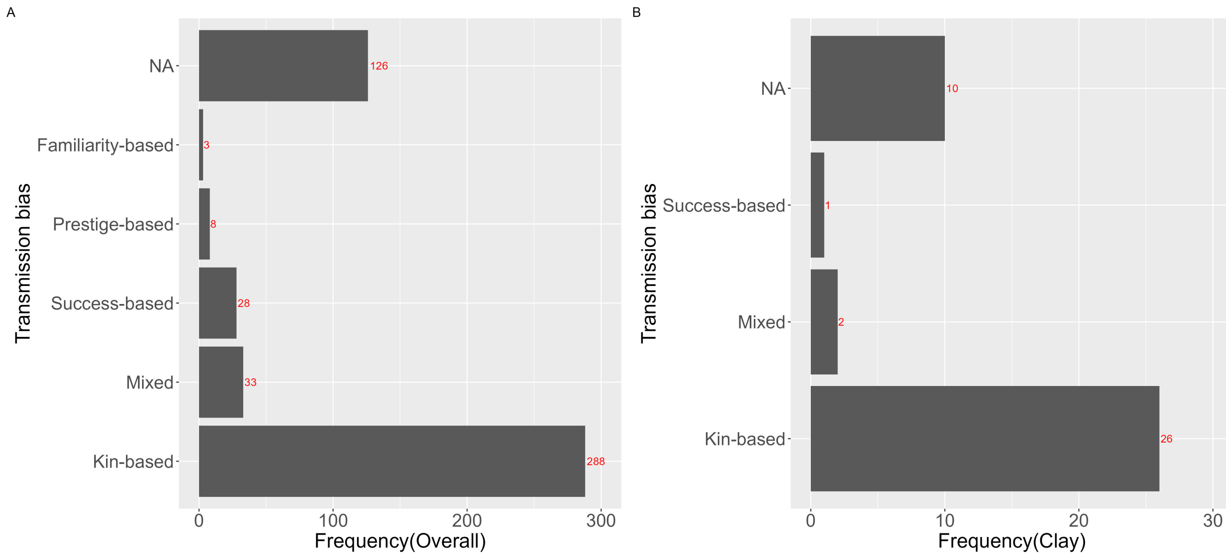

Figure 7: Transmission biases of toolmaking skill acquisition documented in ethnographic texts.

3.2 Missing pattern analysis

The missing data heatmap (**Figure** 8) visualizes the distribution of missing values across observations and variables. High proportions of missing data are evident in variables like Time spent (92%), Location type (87%), Number of learners (87%), Age group of learner (56%), and Number of models (40%). This was further verified by the result of Little's statistical test for missing completely at random (MCAR) data ($p < 0.001$), suggesting this dataset is not missing completely at random.

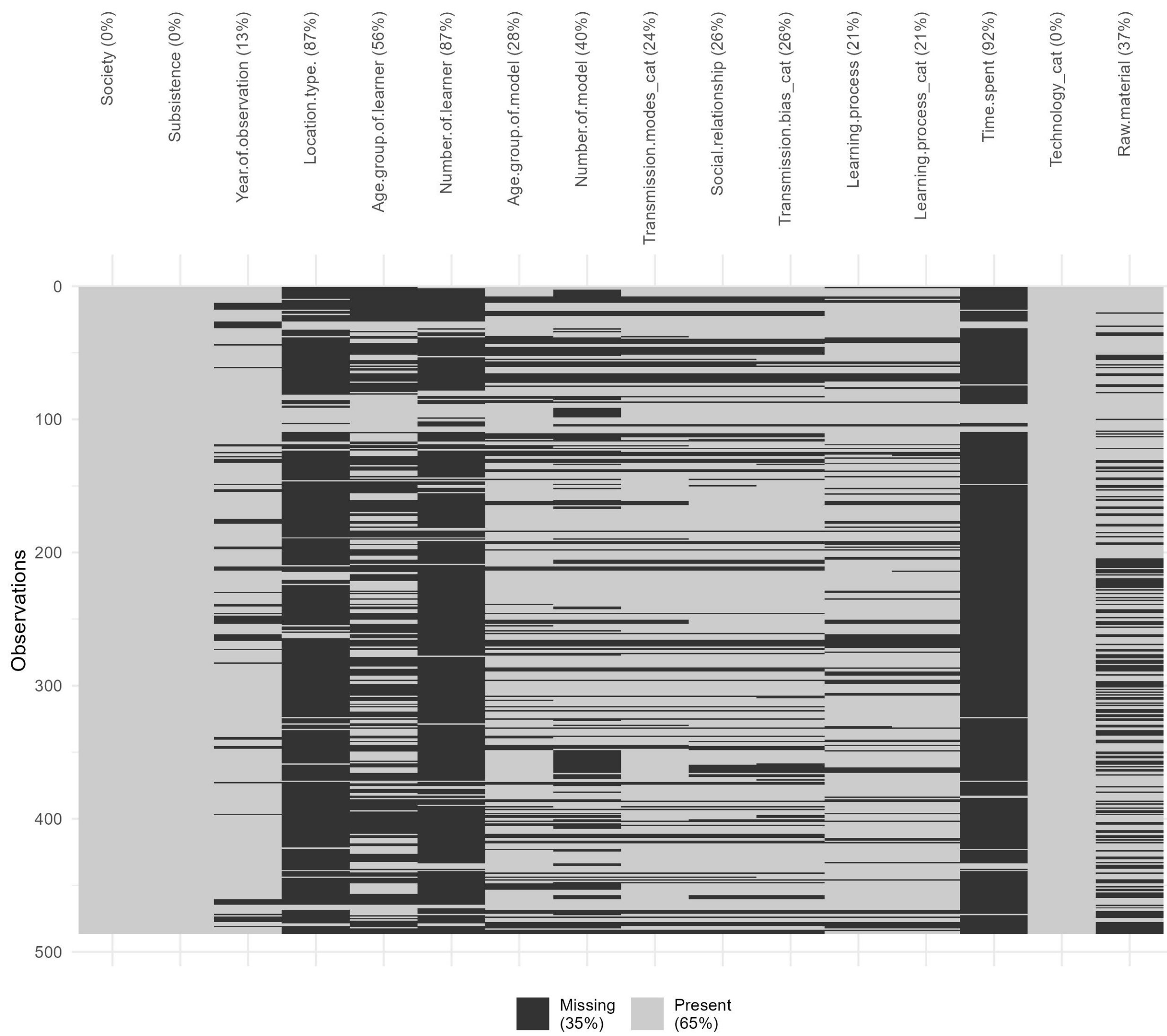

Figure 8: Missing data heatmap. Darker segments represent missing data, while lighter segments indicate available data.

The UpSet plot (**Figure** 9) further illustrates the extent and combinations of missing values among the variables. Notably, 79 observations feature missing data of the four variables with the highest percentage of missing values, including Time spent, number of learners, location type, and age group of learners. In 39 instances, it was found that nine out of ten variables with the highest percentage of missing values are co-missing, and similar patterns occur repeatedly in several intersecting groups. The clustering of missingness across specific variables suggests non-random patterns, possibly due to the varying research interests of the observers as well as the complexity of documenting certain social or learning contexts in the fieldwork.

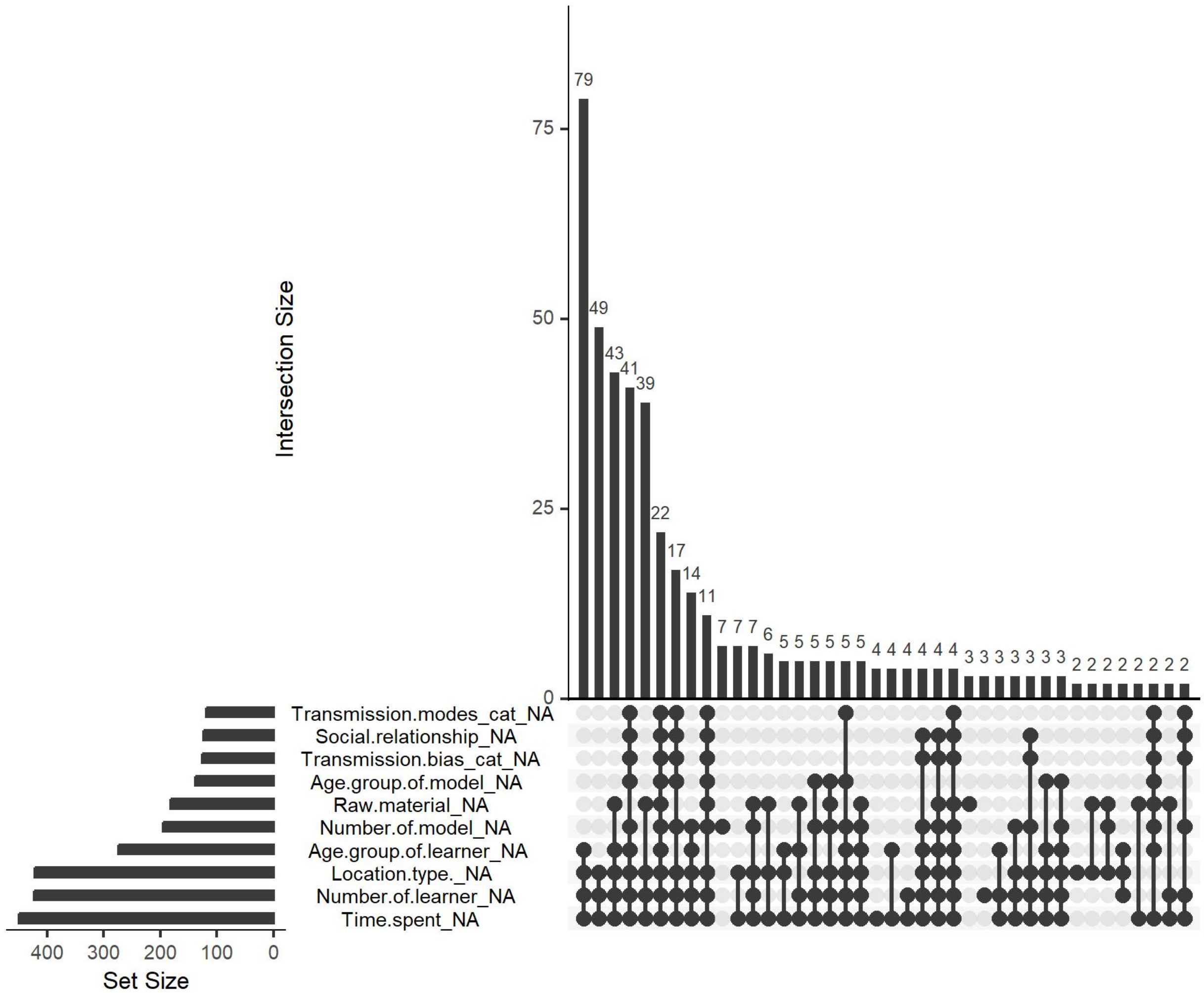

Figure 9: The UpSet plot displaying the co-occurrence of missing variables across observations. Ten variables with the highest percentage of missing values are visualized here.

4 Discussion

While the quantitative results provide a broad overview of the distribution and frequency of documented technologies and raw materials, they do not fully capture the complexity of how these technologies were learned, transmitted, and practiced. During the coding of ethnographic data, unexpected patterns and nuanced insights began to emerge, revealing aspects of technological knowledge and skill that were not initially targeted by pre-designed questions. These qualitative findings offer a deeper contextual understanding that complements the quantitative analysis, highlighting the importance of flexible, data-driven interpretation in uncovering the social dynamics of technology use and production. Following the exact order of the preceding Results section, this discussion will now integrate these qualitative insights with the quantitative findings for each of the six guiding research questions, particularly when, where, how, and why.

4.1 The nature of transmitted skills: Utilitarian crafts and material bias

This uneven distribution could be a mixed result shaped by raw material accessibility and processing difficulty, potential cultural preferences in material use, as well as observational biases. Moreover, the prominence of clothing and basket-making technologies, alongside the frequent use of plant fiber and wood—materials prone to poor preservation—suggests that significant aspects of past material culture are underrepresented in the archaeological record. This highlights the importance of integrating ethnographic data to reconstruct technological practices that are archaeologically invisible, offering a more comprehensive understanding of past human behavior (Desmond, 2022; Gilligan et al., 2024; Langley & Suddendorf, 2020).

4.2 The social fabric of learning: Kinship and peers

These results suggest a strong reliance on a kin-based learning pathway, particularly vertical transmission, for toolmaking skills in non-industrial societies. It provides strong empirical support of Derek and Morgan's (2023) recent claim regarding the importance of vertical channel in skill transmission when compared with knowledge transmission. Taking a closer look at the social relationship between models and learners (see Supplementary Table 1), the author also found that the majority of cases under the categories of oblique transmission involve grandparents, aunts, or uncles, who are also close kin members of the learner. This pattern may reflect the importance of familial and inter-generational knowledge transfer in maintaining craft traditions. Interestingly, the data show limited evidence for peer-to-peer (horizontal) transmission contrary to what has been reported by Lew-Levy and her colleagues (Lew-Levy, Kissler, et al., 2020; Lew-Levy et al., 2023; Lew-Levy & Amir, 2024). It is possible that peer play activities or enculturation through interactions with older siblings falling into a similar age range are not that vital in the domain of toolmaking skills as compared with knowledge domains such as social and gender norms (Lew-Levy et al., 2018), considering the fact that several skills either require many years' practice to master (Bock, 2002; Gurven & Kaplan, 2006; Kaplan et al., 2003) and/or a certain degree of physical strength and motor control that is only available during or after adolescence (Bird & Bliege Bird, 2002; Bliege Bird & Bird, 2002; Jones & Marlowe, 2002). On the other hand, the low frequency of peer-to-peer transmission in the data may not reflect its actual occurrence, but rather the limitations in how ethnographers recognized and recorded learning episodes among children (Hirschfeld, 1999, 2002; Sterelny, 2021). Many brief or informal peer learning interactions, particularly when they were practiced by minors, might have gone undocumented because they did not match Western concepts of teaching and learning, an observational bias that Hewlett and others have noted in ethnographic research (Boyette & Hewlett, 2017; Hewlett & Roulette, 2016).

4.3 Spatial differentiation and task dependency

This pattern suggests that the physical setting of skill transmission is potentially tied to the nature of the craft. Activities involving fine, detailed work or those requiring specialized tools and materials are more often learned indoors, whereas tasks demanding larger spaces or environmental resources, such as boat or trap making, are typically learned outdoors. However, it is also possible that this distribution might simply reflect observers' biases given the size of unreported cases.

While in most cases spatial contexts of toolmaking acquisition activities are under-documented at best, there are sparse ethnographic narratives illustrating the spatial organization of complex skill acquisition like ground stone toolmaking and its relationship with natural resource distribution. For example, Cook (1968: 189-191) noted that different Zapotec villages have different ways of teaching the craft skill of metate and mano:

> "In San Sebastian the finishing tasks in metate and mano production are taught in the residence lot; all preliminary and intermediate production tasks are taught and learned on location in the quarries. In San Juan, on the other hand, where the quarries are close to the village and more accessible than those in San Sebastian, metate size blocks of stone (trozos) are often carted from the quarry to the metatero's residence lot in the village. Thus, the residence lot is, in many instances, the San Juan metatero's workshop."

This paragraph suggests that different toolmaking tasks are learned in distinct contexts, with raw material procurement typically occurring in quarries and subsequent processing tasks taking place in residential areas, depending on the distance between the quarry and the settlement. It suggests that evidence of learning and skill transmission may be spatially segmented, with early-stage production debris concentrated at resource extraction sites (e.g., Castañeda, 2018) and more refined production traces located in domestic settings (e.g., Knight, 2017). Archaeologists must therefore consider the spatial organization of artifact assemblages to identify stages of skill acquisition and social learning. Overlooking this spatial variability could lead to incomplete interpretations of how technical knowledge was transmitted.

4.4 From middle childhood to protracted learning trajectories

Although the clay-related skills might be introduced somewhat earlier in development than the average across all technologies in the dataset, the general distribution suggests that the transmission of tool-making knowledge in non-industrial societies occurs primarily during middle childhood, a critical period in developmental psychology featuring multiple cognitive milestones such as advanced episodic memory (Ghetti & Bunge, 2012), theory of mind (Osterhaus & Koerber, 2021), and executive function (Wilson et al., 2018). Again, the substantial number of unspecified age cases (n=274) indicates limitations in the ethnographic documentation of age-specific learning processes, however, the qualitative data complements this finding, showing the significant role of protracted learning in the formation of human culture (Dallos, 2021).

Several eHRAF ethnographic sources suggested that the learning of toolmaking skills spans multiple developmental periods in an almost structural manner, where each period feature different approaches to learning and varied task priorities. This applies to scenarios where people learn different domains of technology at different ages as well as people acquiring one toolmaking skill learn various technological components in a developmentally scaffolding manner. For instance, in Lancy's (1996: 161-162) classical ethnography on Kpelle children's development, he noted that:

> "Children learn three types of tasks at three different ages. They begin performing such rudimentary tasks of rice farming as cooking, pounding rice, and chasing away birds at a very early age, usually before they are 9. More difficult tasks, such as hoeing on the farm, net weaving, and trap making, are begun around age 9 and mastered

by age 13. Rarer, more complex skills such as cloth weaving, bag weaving, blacksmithing, and medicine making are acquired in the late teens and may be learned after marriage."

Canoe making, a skill that demands not only the acquiring of technical know-how but also the development of proper physical strength to handle heavy logs, best illustrates the case of a long training period required for a single toolmaking skill. For example, Wilbert (1976: 360) commented that among the Warao hunter-gatherers' skill training for men lasted until 25 years old and he further (Wilbert, 1977: 23) wrote that:

> "A boy is associated with canoe-making from early childhood on, initially as spectator, when he accompanies his father and the other men of the family to the forest and watches the production process. Later, as an adolescent, he picks up the axe himself and participates in the process according to his increasing dexterity. By the time he gets married, a young man commands the basic skills and may even have a small canoe to call his own."

This pattern of prolonged and staged learning [c.f. Ifugao weaving; Lambrecht (1958): 1] has also recurrently occurred in other ethnographic observations on other indigenous communities such as Chuuk/Truk (Goodenough, 1951: 53-55) in Micronesia, as well as Kapauku (Pospisil, 1958: 40-41) and Trobriands(Austen, 1945: 194-195; Scoditti, 1989: 56) in Melanesia (e.g., "15-20 years of apprenticeship"). However, given the inherent constraint that most ethnographic sources used here only provided a fragmented narrative on craft learning, it is still unclear how common is this mode of protracted learning in the manufacturing of smaller-sized tools that does not necessarily require a high level of physical strength as debated in several empirical studies regarding humans' unique life history pattern (Bliege Bird & Bird, 2002; Jones & Marlowe, 2002).

During this extended period of learning, the rate at which toolmaking skills are acquired is not constant but instead varies over time, where the initiation ceremony stands out as a period featuring intensive and focused training along with other motor and social skills required for adulthood. This is a recurring pattern that has been widely documented across multiple societies with diverse subsistence strategies, including Laguru (Hamdani, 2001: 11-12), Mende (Little, 1951: 120-121), Eyak (Birket-Smith & Laguna, 1938: 156-157), Kaska (Teit, 1956: 115-119), Pomo (Theodoratus, 1971: 343-344), Malekula (Deacon, 1934: 250-253), Tukano (Hugh-Jones, 1979: 86-87), Goajiro (Watson-Franke & Watson, 1986: 146-147), and Ona (Gusinde & Schütze, 1931: 1451-1452). For example, Hamdani (2001: 11-12) described the female initiation rites in Luguru, East Africa.

> "The 'mwali' initiation rite... She explained that she literally lived in this hut for a period of three years... It was her mother who trained her to sort out green vegetables, to pound the maize and to prepare food. It was her aunts who taught her how to make sleeping mats, baskets, pottery and even caps. Mama Fulora explained that during her grandmother's times, girls used to learn how to make clothing from barks of trees."

Similarly, Hugh-Jones (1979: 86-87) provides a detailed narrative regarding the importance of initiation ceremony in toolmaking skill acquisition with the example Tukano people.

> "For the initiates, the marginal period is a time of education... Then the elders systematically teach the initiates to make all the different kinds of basketry (carrying baskets, sieves, flat trays, cassava squeezers, etc.)... In addition to basketry, the initiates are taught how to make the feather head-dresses and other ritual ornaments used in dances... Just before the blowing of pepper, the initiates are taught to weave conical fish traps which they go and set in streams in the forest, accompanied by their masori."

4.5 Dissecting the complexity of learning processes

The descriptive results indicate that the transmission of tool-making knowledge in non-industrial societies relies heavily on social interaction, be it verbal or gestural, with observation serving as a secondary but significant learning method. The relatively low frequency of individual learning may reflect the observer's bias towards social activities as opposed to following individuals as mentioned in the introduction, which by no means suggests that individual trial-and-error learning is not important in the acquisition of toolmaking skills. Rather, given the inclusive nature between the three categories in this coding scheme, this result suggests that practice is integral to the learning process but pure individual practice without social interaction is rarely documented.

4.5.1 Teaching and causal knowledge in the craftsmanship

This reliance on interaction provides insights into the longstanding debates about the nature and prevalence of teaching across cultures. Teaching and learning almost always appear together in modern education contexts. However, these two concepts are not exactly symmetrical in the sense that learning has a much broader connotation and is widely believed to present in an extensive range of species in the animal kingdom, while teaching is not. Or put differently, learning does not necessarily require the presence of a teacher, while teaching, in its narrowest sense, implies the existence of a learner, even if it is just an imaginative one. In fact, depending on the definition one adopts, it has been long debated whether teaching is a cross-cultural or even cross-species phenomenon (Boyette & Hewlett, 2018). There have been several competing ideas around the presence and scope of teaching in human societies. The first position is represented by classical cultural and psychological anthropologists like Mead (1964) and Lancy (2010) who adopt a very narrow definition of teaching, arguing that teaching does not exist or is rare in small-scale cultures. Lancy (2010: 97) believed teacher is "nuanced, student-centered, developmentally appropriate instruction by dedicated adults" and is a recent product of "a long process of education changes".

On the other hand, Hewlett and Roulette (2016) took a minimalist approach and defined teaching as "an individual modifies her/his behaviour to enhance learning in another". Under this definition, teaching was identified through video recordings during as early as the infancy period in a variety of forms among the Aka hunter-gatherers of the Central African Republic and Republic of Congo, ranging from natural pedagogy to opportunity scaffolding to verbal instruction. Their research suggested that most teaching behaviors documented in the Aka foragers are usually very short and subtle, lasting only a few seconds but occurring repeatedly. The combination of short duration and observational subtlety of these types of teaching behaviors provide one possible explanation for why teaching was believed to be absent or rare for early anthropologists, in addition to the debates around the definition.

As observed in several non-industrial societies, including Kaska (Honigmann & Bennett, 1949: 185-186), Warao (Wilbert, 1976: 318-319), Kpelle (Lancy, 1975: 151), Haida (Blackman & Davidson, 1982: 182), and Tiv (Bohannan, 1958: 381), explicit transmission of causal knowledge appears to be rarely documented in non-industrial societies. Below is a direct quote from Honigmann (1949: 185-186), who provides positive evidence of the missing among Kaska foragers living in the Arctic region.

> "Education. Most education for living is received by the child in the course of performing routine activities. Such education is unformalized and undirected... Sometimes generalized instructions are furnished to children who, in carrying these out, often become confused. Thus we observed one fourteen year old girl make tea by putting the leaves into cold water. Nobody referred to the muddy quality of the beverage. The same kind of teaching was used with the ethnographer. Activities were illustrated but never explained in detail, the exposition of principles being ignored."

These materials provide interesting insights into the longstanding debates about the nature and prevalence of teaching across cultures. If teaching is narrowly defined as the deliberate and structured transmission of explicit causal knowledge, which is essentially a Euro-centric definition, then its scarcity supports earlier claims by scholars like Mead (1964) and Lancy (2010) that teaching is very limited in small-scale societies. In these contexts, learning often occurs through observation, imitation, and participation rather than through formal instruction. As suggested by Hewlett and Roulette (2016), teaching in non-industrial societies may be subtle, brief, and embedded within daily activities, making it less visible to researchers employing narrow definitions. Therefore, recognizing the diversity of teaching practices requires adopting broader, more inclusive definitions that account for indirect and informal forms of knowledge transmission, such as the TEACH ethogram developed by Kline (2017).

Additionally, these findings contribute to ongoing discussions in cultural evolutionary research, where experimental (Derex et al., 2019) and ethnographic studies (Harris et al., 2021) have suggested that explicit causal understanding is not necessary for the transmission of complex cultural traits (cf. Högberg et al., 2024; Osiurak et al., 2021; Osiurak & Reynaud, 2020). It should be first clarified that causal knowledge here refers to exclusively the mechanical and physical knowledge regarding how materials interact with each other at the very basic level, which is different from causal perception that is prevalent in all kinds of indigenous ontologies. The recurring positive evidence of the lack of causal knowledge transmission demonstrates in this project that individuals in non-industrial societies can effectively learn and transmit toolmaking skills without formal instruction in causal mechanisms, which means high-fidelity cultural transmission can occur through embodied practice and situated learning rather than explicit teaching. Recognizing that causal knowledge is not necessarily a prerequisite for cultural continuity (Derex et al., 2019; Harris et al., 2021) challenges assumptions about the role of explicit reasoning in technological advancement and highlights the importance of contextual, experience-based learning in sustaining complex cultural traditions. It should be noted that the author does not try to make an argument here that causal reasoning is completely absent in the minds of craftsmen in non-industrial societies. Instead, the transmission of tacit knowledge like toolmaking skills seldom revolves around the specification of causal mechanisms behind it.

4.5.2 Alternative pedagogies: Co-production and miniature model

Instead of providing explicitly step-by-step causal knowledge-oriented teaching, a common way of acquiring a toolmaking skill is that experienced models co-produce an object with novices. When a novice makes mistakes, the experienced model takes it back and corrects the mistake and then returns it to the novice again for them to finish. During this process, no or little verbal explanation regarding why a specific step is necessary or how to avoid similar mistakes in the future is provided, as vividly described in Lancy's (1975: 151) childhood-focused dissertation fieldwork in Kpelle horticulturalists in Western Africa.

> "Wollokollie's youngest wife, Goma was an expert at Bii paa or bag weaving... Now she is teaching a teenaged girl of 16 how to do it. The girl has prepared the strands of palm leaf and brought them to Goma to help her get started. The two alternate at weaving the bag. The girl makes a mistake and Goma unravels what she has done, does it correctly for awhile, then lets the girl take over again. Goma can't explain how to do it, either to me or the girl. She can only show and guide our hands as I, too, make an attempt to learn."

On top of this Kpelle bag weaving case, similar social interaction dynamics have also been observed among Garo people in South Asia regarding how the skill of basket making is transmitted between male models and learners (Burling, 1963: 116).

> “Only after he is about twelve years old does anything like formal teaching take place. The most difficult skill to learn or to teach is basket-making, which the father or some other man must impart to a boy. The instruction consists mostly of telling a child to make a particular basket as he has watched others do and then reprimanding him if he does it wrong. Occasionally a man will demonstrate, but I never saw anyone who was able to analyze and point out the mistakes another was making. If a person makes a mistake in a complex basket, for instance, his instructor will take the basket away, redo it himself, and then tell him to do it that way henceforth. No effort is made to break the complex weaving process down into steps so as to make it easy to understand and learn systematically, although it is recognized that certain baskets are easier to weave than others. Fathers tell their sons to make the easier types first, and then gradually lead them to the more difficult ones.”

This recurring phenomenon, as also documented in Shona (Aschwanden, 1982: 62), Tallensi (Fortes, 1938 : 60-61), Tzeltal (Nash, 1970: 274-275), Gros Ventre (Flannery, 1953: 159-160), Ifugao (Barton, 1938: 43) and many other societies, has some deep implications for archaeologists who are interested in identifying the skill level of producers based on certain morphological traits of individual specimen or a whole artifact assemblage. In this pedagogical scenario, the experienced individual and novice co-produce an item, and part or all of the mistakes made by novices can be erased by the former’s behaviors. The coexistence of fossilized actions of both stakeholders preserved in archaeological materials often provides us a precious opportunity to reconstruct the complex learning behaviors in the past, be it stone artifacts (Assaf et al., 2016; Assaf, 2019) or pottery vessels (Crown, 2007).

Another commonly used pedagogical approach without resorting to explicit causal knowledge transmission is the making of miniature models for and by children underage. To accommodate children’s developmental constraints in hand size and physical strength (Kilgore et al., 2025), more experienced individuals make smaller and often simplified versions of the target artifact and then pass them to children as models to begin with. It has been observed in many different societies and technological domains, including Blackfoot (Ewers, 1958: 103) bow and arrow making, Trobriands (Weiner, 1976: 192) net making, Jivaro (Harner, 1973: 92-93) pottery vessels and figurines, Mbuti carrying basket making (Turnbull, 1965: 125-126), etc. In terms of their implication for material culture research, these miniature models possess interesting properties of serving both as toys and tools (Lew-Levy et al., 2022; Riede et al., 2023), and their producers can be either adults or children.

4.6 Why this model? Why learning at all?

The high frequency of kin-based transmission in our data aligns with previous research on traditional skill acquisition such as the kin-based adze making system observed in West Papua (Perequin & Petrequin, 2020; Stout, 2002) as well as the Maya weaving apprenticeship across multiple generations in domestic settings (Greenfield et al., 2003; Maynard et al., 2024). This finding is particularly interesting because traditionally the field of cultural evolutionary model focuses mainly on success/prestige bias or conformist bias (copying the majority), while kin-based bias remains understudied. A potential explanation is that success/prestige bias (Chellappoo, 2021; Chudek et al., 2012; Jiménez & Mesoudi, 2019) or conformist bias (Heinrich Mora et al., 2025; Henrich & Boyd, 1998; McGuigan et al., 2012; Wakano & Aoki, 2007) may be more important in other cultural domains like the transmission of beliefs or ascetic preferences, while toolmaking usually is a long-term learning process featuring a more intimate social relationship. This pattern, combined with the argument made by Henrich (2020) that kinship is central to everyone but the WEIRD population, suggested the much-needed empirical recalibration to cultural evolution modeling regarding the research priority of kin-based transmission bias.

4.6.1 Distribution of expertise

This dominance of kinship is further contextualized by the sparse distribution of expertise in many non-industrial societies. In many non-industrial societies, while basic toolmaking skills are widely shared, especially plant-based technologies, individuals explicitly recognized as experts in ethnographic description are relatively rare. This sparse distribution pattern of expertise within a given community can be usually attributed to two key factors. First, the role of innate talent or individual aptitude in motor skills is often emphasized, as seen among the Nuba (Nadel, 1947: 71), Bemba (Richards, 1939: 105-106), Yukaghir (Jochelson, 1910: 425), Malay (Wilkinson, 1920: 12-13), and Southern Coast Salish (Smith, 1940: 305-306), where certain individuals are believed to possess natural abilities that set them apart in craftsmanship. On the contrary, those who do not show success in their initial attempts often are not encouraged to continue the efforts. For example, in his field observation among Nuba people in Sudan, British anthropologist Fred Nadel (1947: 71) noted that:

> "Rather, craftsmanship appears irregularly, as a result of the inclinations and the cleverness of certain individuals which, once they become known, are utilized by others who are not so 'clever with their fingers'. Everybody can, in theory, make baskets or beer-strainers; but in most groups we find 'experts' to whom people would go for these things in preference to making them (much less satisfactorily) themselves. The same is true of the manufacture of native stools, angrebs, and certain musical instruments like the Nuba lyre or wooden drums (the skin is fixed by the owner himself). The list of self-produced goods, on the other hand, comprises the old (wooden) farming tools and spears, shields, sticks, the handles of hoes and axes, the old-type plank-bed, simpler musical instruments like flutes, animal horns, or gourd trumpets, and, finally, the native house, complete with roof and granaries."

Second, highly specialized crafts featuring rare/expensive raw materials combined with small community size often lead to a low market demand, which can further limit the development of expertise. For example, among the Malays, Wilkinson noted that (1920: 12-13) noted that "a village could not support two smiths", where "the more skilful artificer soon drove out his rival and monopolised what work there was." In this instance, the small settlement and population size constrained widespread mastery of the craft. Interestingly, when a community is large enough to support multiple silversmiths as in the case of Navajo (Adair, 1944: 91), they "do not like other Navajo to watch them while they are working because they do not want their art stolen", developing a sense of intellectual property. These factors, together with the strong constraining influence of kinship in determining who can access knowledge and materials, help explain the scarcity of clear cases of pure success-based transmission bias found in the analysis of the "why" question.

4.6.2 Skill training for community integration

Beyond the choice of why to learn from a specific model, the qualitative data also addresses why these skills are learned at all. Several sources in this study indicate that local communities perceive toolmaking not merely as a survival skill but as a vital process for social integration.

Below are two short examples respectively from Tzeltal (Nash, 1970: 321) and Kpelle (Lancy, 1996: 154-155) illustrating toolmaking skill acquisition as a process of socialization.

> "Learning agricultural tasks and pottery-making in the household is part of adjusting to membership in the community. Parents are responsible for teaching the value of industriousness and are held accountable throughout the life of the child. Agricultural tasks are scheduled in a yearly cycle that provides sufficient flexibility for a man to carry out community and familial ceremonial responsibilities."

> "In each of the previous cases, we see individuals acquiring weaving skills to enhance their own social participation. As Sua indicates, if a girl wants to"look nice," to match the appearance of women, she must learn the skills to dress and decorate herself. In attempting to do so, she will be accorded the status of a "legitimate peripheral participant." She may be supplied with materials and someone competent may agree to model the skill and also to critique the fledgling product. Miniature tools may be supplied."

This perspective fits into Lave and Wenger's (1998) famous concept of "communities of practice," which was originally developed to theorize industrialized contexts but have broader relevance to the non-industrialized societies we described here. In these communities of practice, learning is inherently social and occurs through active participation in shared cultural practices, where engaging in toolmaking allows individuals, particularly younger members, to become integrated into the social fabric of their community. Through observation, imitation, and guided practice, learners not only acquire technical skills but also internalize social norms, values, and identities associated with craftsmanship. This participation fosters a sense of belonging and reinforces communal bonds, demonstrating that toolmaking serves both practical and social functions. Thus, tool production becomes a medium through which individuals transition from peripheral participation to full membership within their cultural group, highlighting the dual role of skill transmission in sustaining both material needs and social cohesion.

5 Conclusion

Back to the six basic questions regarding toolmaking skill acquisition in non-industrial societies proposed in the introduction, here the author draws on key findings from both quantitative analyses and qualitative insights to provide brief answers as follows.

What: The two most frequent toolmaking skills are clothes making and basket making, documented in many non-industrial societies regarding subsistence strategies or geographical distribution. From the perspective of raw material exploitation, prevalent yet perishable materials like plant fiber and wood are the dominant categories.

Who: People learn most toolmaking skills from their close kin members, particularly parents. Extended family members like grandparents and uncles/aunts are also heavily involved in this process.

Where: Indoor learning scenarios are more frequently documented compared with the outdoor ones. The former is often associated with tasks involving meticulous motor control and specialized tools and materials, while activities demanding larger spaces or environmental resources are typically learned outdoors. For complex crafts involving multiple steps like raw material procurement, the locations are dependent on the specific task involved.

When: For many skills, they started the learning processes in middle childhood (6-12 years old), which can last for several years. The initiation ceremonies provide opportunities for a period of intensive training.

How: Although individual practice and trial and error always exist, toolmaking skill acquisition involves lots of social interactions, during which causal knowledge is rarely explicitly transmitted. Instead, two commonly used pedagogical approaches are 1) object co-production between experienced and inexperienced individuals as well as 2) miniature models manufactured for and by underage novices.

Why: This issue can be interpreted from two different angles, namely why people in non-industrial societies learn toolmaking skills and why people choose to learn them from certain individuals. For the former question, toolmaking skill transmission is regarded as an important way of community integration on top of survival needs. For the latter question, kinship is the main (constraining) factor when choosing a model rather than expertise, prestige, or direct payoff comparison.

This project represents the most extensive compilation of its kind to date, offering unparalleled breadth and depth in documenting cultural transmission across diverse crafts and societies. Its comprehensive scope provides a valuable foundation for exploring patterns and dynamics in cultural learning and technological practices, enabling analyses that were previously unattainable due to limited data availability. The qualitative analysis provides interesting insights into some major theoretical issues in cross-cultural studies such as the scope of teaching in hunter-gatherer societies, the role of explicit causal knowledge in cultural evolution, community of practice, etc.

Regarding its archaeological implications, the clay-specific taphonomic analysis largely mirrors the general patterns observed in the broader dataset, with minor variations and occasional missing categories likely attributable to the smaller sample size rather than substantive differences in transmission mechanisms. As such, this project also serves as a reference framework for researchers interested in inferring past learning dynamics from archaeological records (Liu & Stout, 2023) by refining prior assumptions when fine-grained and contextualized empirical data are lacking, such as favoring kin-based transmission over random copying when formulating initial hypotheses on transmission bias. Moreover, these findings can guide the design of future experimental archaeology research by providing a baseline for more realistic social learning scenarios (Liu, 2024). For example, experimental conditions can be designed to reflect kin-based social structures instead of pairings of strangers. Researchers can also incorporate the interactive teaching and error correction that this study shows are central to skill acquisition.

However, the presence of substantial and systematically patterned missing data introduces important limitations, which is not uncommon for cross-cultural datasets. The analysis indicates that missing values are not randomly distributed but are likely missing at random (MAR) or missing not at random (MNAR). This non-random missingness necessitates caution in conducting further statistical analyses, as it may bias results if not properly addressed. Careful application of imputation methods and sensitivity analyses will be essential to ensure the robustness and reliability of future findings derived from this dataset. In the end, the author sees it as a stepping stone instead of a definitive text for a better understanding of the subject matter, which merely serves as a starting point for an open dialogue and a project to be collaborated with future researchers like the "Play Object Play" database (Riede et al., 2023).

6 Acknowledgment

This work was assisted in part by skills learned at the 2022 NSF HRAF Summer Institute for Cross Cultural Anthropological Research (NSF BCS Grant #2020156 to HRAF). The revision of this manuscript was funded by the European Research Council Synergy Grant for the project Evolution of Cognitive Tools for Quantification (QUANTA), No. 951388. I would like to thank Carol Ember, Fiana Jordan, and Sean Roberts for their suggestions on the research design of this project. I am also grateful to Dietrich Stout, Craig Hadley, Claudio Tennie and two anonymous reviewers for their constructive comments on an earlier draft of this article.

References

Adair, J. (1944). *The Navajo and Pueblo Silversmiths.* University of Oklahoma Press.

Aschwanden, H. (1982). *Symbols of Life: An Analysis of the Consciousness of the Karanga.* Mambo Press.

Assaf, E. (2019). Core sharing: The transmission of knowledge of stone tool knapping in the Lower Palaeolithic, Qesem Cave (Israel). *Hunter Gatherer Research*, *3*(3), 367–399. https://doi.org/10.3828/hgr.2017.19

Assaf, E., Barkai, R., & Gopher, A. (2016). Knowledge transmission and apprentice flint-knappers in the Acheulo-Yabrudian: A case study from Qesem Cave, Israel. *Quaternary International*, *398*, 70–85. https://doi.org/10.1016/j.quaint.2015.02.028

Austen, L. (1945). New Guinea: Material Culture.: Native Handicrafts in the Trobriand Islands. *Mankind*, *3*(7), 193–198. https://doi.org/10.1111/j.1835-9310.1945.tb01306.x

Barton, R. F. (1938). *Philippine Pagans: The Autographies of Ifugaos.* George Routledge & Sons.

Becker, S. O., & Woessmann, L. (2024). Education and Socioeconomic Development During the Industrialization. In C. Diebolt & M. Haupert (Eds.), *Handbook of Cliometrics* (pp. 459–479). Springer International Publishing. https://doi.org/10.1007/978-3-031-35583-7_75

Bira, T. G., & Hewlett, B. S. (2023). Cultural Learning Among Pastoralist Children. *Cross-Cultural Research*, *57*(1), 74–110. https://doi.org/10.1177/10693971221134180

Birch, J. (2021). Toolmaking and the evolution of normative cognition. *Biology & Philosophy*, *36*(1), 4. https://doi.org/10.1007/s10539-020-09777-9

Bird, D. W., Bird, R. B., Codding, B. F., & Zeanah, D. W. (2019). Variability in the organization and size of hunter-gatherer groups: Foragers do not live in small-scale societies. *Journal of Human Evolution*, *131*, 96–108. https://doi.org/10.1016/j.jhevol.2019.03.005

Bird, D. W., & Bliege Bird, R. (2002). Children on the reef. *Human Nature*, *13*(2), 269–297. https://doi.org/10.1007/s12110-002-1010-9

Birket-Smith, K., & Laguna, F. D. (1938). *The Eyak Indians of the Copper River Delta, Alaska*. Levin & Munksgaard(E. Munksgaard).

Blackman, M. B., & Davidson, F. E. (1982). *During My Time: Florence Edenshaw Davidson, a Haida Woman*. University of Washington Press.

Bliege Bird, R., & Bird, D. W. (2002). Constraints of knowing or constraints of growing? : Fishing and collecting by the children of mer. *Human Nature*, *13*(2), 239–267. https://doi.org/10.1007/s12110-002-1009-2

Bock, J. (2002). Learning, life history, and productivity. *Human Nature*, *13*(2), 161–197. https://doi.org/10.1007/s12110-002-1007-4

Bohannan, P. (1958). *Three Source Notebooks in Tiv Ethnography: A source notebook in the Tiv life cycle*. Human Relations Area Files.

Boyette, A. H., & Hewlett, B. S. (2017). Autonomy, Equality, and Teaching among Aka Foragers and Ngandu Farmers of the Congo Basin. *Human Nature*, *28*(3), 289–322. https://doi.org/10.1007/s12110-017-9294-y

Boyette, A. H., & Hewlett, B. S. (2018). Teaching in Hunter-Gatherers. *Review of Philosophy and Psychology*, *9*(4), 771–797. https://doi.org/10.1007/s13164-017-0347-2

Buckley, C. (2023). Craft Traditions. In J. J. Tehrani, J. Kendal, & R. Kendal (Eds.), *The Oxford Handbook of Cultural Evolution*. Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198869252.013.40

Burling, R. (1963). *Rengsanggri: Family and Kinship in a Garo Village*. University of Pennsylvania Press.

Castañeda, N. (2018). Apprenticeship in early neolithic societies: The transmission of technological knowledge at the flint mine of casa montero (madrid, spain), ca. 5300–5200 cal BC. *Current Anthropology*, *59*(6), 716–740. https://doi.org/10.1086/700837

Chellappoo, A. (2021). Rethinking prestige bias. *Synthese*, *198*(9), 8191–8212. https://doi.org/10.1007/s11229-020-02565-8

Chudek, M., Heller, S., Birch, S., & Henrich, J. (2012). Prestige-biased cultural learning: Bystander's differential attention to potential models influences children's learning. *Evolution* *and Human Behavior*, *33*(1), 46–56. https://doi.org/10.1016/j.evolhumbehav.2011.05.005

Cook, S. (1968). *Teitipac and its Metateros: An economic anthropological study of production and exchange in a peasant-artisan economy in the valley of Oaxaca, Mexico* [Doctoral dissertation]. University of Pittsburgh.

Coon, C. S. (1931). *Tribes of the Rif*. Peabody museum of Harvard university.

Crown, P. L. (2007). Life Histories of Pots and Potters: Situating the Individual in Archaeology. *American Antiquity*, *72*(4), 677–690. https://doi.org/10.2307/25470440

Dallos, C. (2021). Is there more to human social learning than enhanced facilitation? Prolonged learning and its impact on culture. *Humanities and Social Sciences Communications*, *8*(1), 1–10. https://doi.org/10.1057/s41599-021-00829-3

Deacon, A. B. (1934). *Malekula: A vanishing people in the New Hebrides*. G. Routledge & Sons. http://archive.org/details/malekulavanishin0000arth

Derex, M., Bonnefon, J.-F., Boyd, R., & Mesoudi, A. (2019). Causal understanding is not necessary for the improvement of culturally evolving technology. *Nature Human Behaviour*, *3*(5), 446–452. https://doi.org/10.1038/s41562-019-0567-9

Derex, M., & Morgan, T. J. H. (2023). The Cultural Transmission of Technological Skills. In J. J. Tehrani, J. Kendal, & R. Kendal (Eds.), *The Oxford Handbook of Cultural Evolution*. Oxford University Press. https://doi.org/10.1093/oxfordhb/9780198869252.013.32

Desmond, A. (2022). Bone tool proxy evidence for coiled basketry production in the north african palaeolithic. *Journal of African Archaeology*, *20*(2), 156175. https://brill.com/view/journals/jaa/20/2/article-p156_2.xml

Dunmore, C. J., Karakostis, F. A., Leeuwen, T. van, Lu, S.-C., & Proffitt, T. (2023). Chapter 6 - Tool use and the hand. In C. S. Hirst, R. J. Gilmour, K. A. Plomp, & F. A. Cardoso (Eds.), *Behaviour in Our Bones* (pp. 135–171). Elsevier. https://doi.org/10.1016/B978-0-12-821383-4.00011-5

Ewers, J. C. (1958). *The Blackfeet: Raiders on the Northwestern Plains*. University of Oklahoma Press.

Flannery, R. (1953). *The Gros Ventres of Montana*. Catholic University of America Press.

Fortes, M. (1938). *Social and psychological aspects of education in Taleland*. Oxford University Press. http://archive.org/details/socialpsychologi00fort

Garfield, Z. H., Garfield, M. J., & Hewlett, B. S. (2016). A Cross-Cultural Analysis of Hunter-Gatherer Social Learning. In H. Terashima & B. S. Hewlett (Eds.), *Social Learning and Innovation in Contemporary Hunter-Gatherers: Evolutionary and Ethnographic Perspectives* (pp.

19–34). Springer Japan. https://doi.org/10.1007/978-4-431-55997-9_2

Gerring, J. (2012). Mere Description. *British Journal of Political Science*, *42*(4), 721–746. https://doi.org/10.1017/S0007123412000130

Ghetti, S., & Bunge, S. A. (2012). Neural changes underlying the development of episodic memory during middle childhood. *Developmental Cognitive Neuroscience*, *2*(4), 381–395. https://doi.org/10.1016/j.dcn.2012.05.002

Gilligan, I., d'Errico, F., Doyon, L., Wang, W., & Kuzmin, Y. V. (2024). Paleolithic eyed needles and the evolution of dress. *Science Advances*, *10*(26), eadp2887. https://doi.org/10.1126/sciadv.adp2887

Goodenough, W. H. (1951). *Property, kin, and community on Truk*. Yale University Press. https://ehrafworldcultures.yale.edu/cultures/or19/documents/001

Greenfield, P. M., Maynard, A. E., & Childs, C. P. (2003). Historical change, cultural learning, and cognitive representation in Zinacantec Maya children. *Cognitive Development*, *18*(4), 455–487. https://doi.org/10.1016/j.cogdev.2003.09.004

Gurven, M., & Kaplan, H. (2006). Determinants of time allocation across the lifespan. *Human Nature*, *17*(1), 1–49. https://doi.org/10.1007/s12110-006-1019-6

Gusinde, M., & Schütze, F. (1931). *The fireland indians: Vol. 1. The selk'nam, on the life and thought of a hunting people of the great island of tierra del fuego*. Verlag der Internationalen Zeitschrift.

Hamdani, S. (2001). Female Adolescent Rites and the Reproductive Health of Young Women in Morogoro, Tanzania. *Bulletin of the International Committee on Urgent Anthropological and Ethnological Research*, *41*, 165–181. https://www.africabib.org/rec.php?RID=W00076788

Harner, M. J. (1973). *The Jivaro: People of the Sacred Waterfalls*. Doubleday Natural History Press.

Harris, J. A., Boyd, R., & Wood, B. M. (2021). The role of causal knowledge in the evolution of traditional technology. *Current Biology*, *31*(8), 1798–1803.e3. https://doi.org/10.1016/j.cub.2021.01.096

Heinrich Mora, E., Denton, K. K., Palmer, M. E., & Feldman, M. W. (2025). Conformity to continuous and discrete ordered traits. *Proceedings of the National Academy of Sciences*, *122*(3), e2417078122. https://doi.org/10.1073/pnas.2417078122

Henrich, J. (2020). *The WEIRDest people in the world: How the west became psychologically peculiar and particularly prosperous*. Farrar, Straus; Giroux.

Henrich, J., & Boyd, R. (1998). The evolution of conformist transmission and the emergence of between-group differences. *Evolution and Human Behavior*, *19*(4), 215–241. https://doi.org/10.1016/S1090-5138(98)00018-X

Hewlett, B. S., Boyette, A. H., Lew-Levy, S., Gallois, S., & Dira, S. J. (2024). Cultural transmission among hunter-gatherers. *Proceedings of the National Academy of Sciences*, *121*(48), e2322883121. https://doi.org/10.1073/pnas.2322883121

Hewlett, B. S., & Roulette, C. J. (2016). Teaching in hunter–gatherer infancy. *Royal Society Open Science*, *3*(1), 150403. https://doi.org/10.1098/rsos.150403

Hirschfeld, L. A. (1999). L'enfant terrible: Anthropology and its aversion to children. *Etnofoor*, *12*(1), 5–26. https://www.jstor.org/stable/25757950

Hirschfeld, L. A. (2002). Why don't anthropologists like children? *American Anthropologist*, *104*(2), 611–627. https://doi.org/10.1525/aa.2002.104.2.611

Högberg, A., Lombard, M., Högberg, A., Iliefski-Janols, E., Lindblad, G., Almér, A., Thompson, W. H., Rost, M., Andreasson, S., Wiig, A., & Gärdenfors, P. (2024). Human sociotechnical evolution through the lens of an abstracted-wheel experiment: A critical look at a micro-society laboratory study. *PLOS ONE*, *19*(11), e0310503. https://doi.org/10.1371/journal.pone.0310503

Honigmann, J. J., & Bennett, W. C. (1949). *Culture and ethos of Kaska society*. Yale University Press. https://ehrafworldcultures.yale.edu/cultures/nd12/documents/001

Hugh-Jones, S. (1979). *The Palm and the Pleiades: Initiation and Cosmology in Northwest Amazonia*. Cambridge University Press.

Hulstaert, G. E. (1938). *Marriage among the nkundu*. Librairie Falk fils.

Jiménez, Á. V., & Mesoudi, A. (2019). Prestige-biased social learning: current evidence and outstanding questions. *Palgrave Communications*, *5*(1), 1–12. https://doi.org/10.1057/s41599-019-0228-7

Jochelson, W. (1910). *The Yukaghir and the Yukaghirized Tungus*. E.J. Brill & G. E. Stechert.

Jones, N. B., & Marlowe, F. W. (2002). Selection for delayed maturity: Does it take 20 years to learn to hunt and gather? *Human Nature*, *13*(2), 199–238. https://doi.org/10.1007/s12110-002-1008-3

Kaplan, H., Lancaster, J., & Robson, A. (2003). Embodied capital and the evolutionary economics of the human life span. *Population and Development Review*, *29*, 152–182. https://www.jstor.org/stable/3401350

Karakostis, F. A., Haeufle, D., Anastopoulou, I., Moraitis, K., Hotz, G., Tourloukis, V., & Harvati, K.

(2021). Biomechanics of the human thumb and the evolution of dexterity. *Current Biology*, *31*(6), 1317–1325.e8. https://doi.org/10.1016/j.cub.2020.12.041

Kendal, R. L., Boogert, N. J., Rendell, L., Laland, K. N., Webster, M., & Jones, P. L. (2018). Social Learning Strategies: Bridge-Building between Fields. *Trends in Cognitive Sciences*, *22*(7), 651–665. https://doi.org/10.1016/j.tics.2018.04.003

Kilgore, M. B., Liu, C., & Stout, D. (2025). What can lithics tell us about the evolution of learning and development? *Archaeometry*. https://doi.org/10.1111/arcm.70002

Kivell, T. L., Baraki, N., Lockwood, V., Williams-Hatala, E. M., & Wood, B. A. (2023). Form, function and evolution of the human hand. *American Journal of Biological Anthropology*, *181*(S76), 6–57. https://doi.org/10.1002/ajpa.24667

Kline, M. A. (2017). Teach: An Ethogram-based Method to Observe and Record Teaching Behavior. *Field Methods*, *29*(3), 205–220. https://doi.org/10.1177/1525822X16669282

Knight, C. L. F. (2017). Children, apprenticeship and pedagogy: Domestic crafting and obsidian core production at the zaragoza-oyameles source area in puebla, mexico. *Journal of Anthropological Archaeology*, *47*, 152–166. https://doi.org/10.1016/j.jaa.2017.02.003

Lambrecht, F. (1958). Ifugaw weaving. *Folklore Studies*, *17*, 1–53. https://doi.org/10.2307/1177377

Lancy, D. F. (1975). *Work, play and learning in a Kpelle town* [Doctoral {Dissertation}, University of Pittsburgh]. https://ehrafworldcultures.yale.edu/cultures/fd06/documents/009

Lancy, D. F. (1996). *Playing on the mother-ground: Cultural routines for children's development*. Guilford Press.

Lancy, D. F. (2010). Learning 'from nobody': The limited role of teaching in folk models of children's development. *Childhood in the Past*, *3*(1), 79–106. https://doi.org/10.1179/cip.2010.3.1.79

Lancy, D. F. (2017). Homo faber juvenalis: A multidisciplinary survey of children as tool makers/users. *Childhood in the Past*, *10*(1), 72–90. https://doi.org/10.1080/17585716.2017.1316010

Langley, M. C., & Suddendorf, T. (2020). Mobile containers in human cognitive evolution studies: Understudied and underrepresented. *Evolutionary Anthropology: Issues, News, and Reviews*, *29*(6), 299–309. https://doi.org/10.1002/evan.21857

Lew-Levy, S., & Amir, D. (2024). Children as agents of cultural adaptation. *Behavioral and Brain Sciences*, 1–68. https://doi.org/10.1017/S0140525X24001377

Lew-Levy, S., Andersen, M. M., Lavi, N., & Riede, F. (2022). Hunter-Gatherer Children's Object Play and Tool Use: An Ethnohistorical Analysis. *Frontiers in Psychology*, *13*, 824983. https://doi.org/10.3389/fpsyg.2022.824983

Lew-Levy, S., Bos, W. van den, Corriveau, K., Dutra, N., Flynn, E., O'Sullivan, E., Pope-Caldwell, S., Rawlings, B., Smolla, M., Xu, J., & Wood, L. (2023). Peer learning and cultural evolution. *Child Development Perspectives*, *17*(2), 97–105. https://doi.org/10.1111/cdep.12482

Lew-Levy, S., Kissler, S. M., Boyette, A. H., Crittenden, A. N., Mabulla, I. A., & Hewlett, B. S. (2020). Who teaches children to forage? Exploring the primacy of child-to-child teaching among Hadza and BaYaka Hunter-Gatherers of Tanzania and Congo. *Evolution and Human Behavior*, *41*(1), 12–22. https://doi.org/10.1016/j.evolhumbehav.2019.07.003

Lew-Levy, S., Lavi, N., Reckin, R., Cristóbal-Azkarate, J., & Ellis-Davies, K. (2018). How Do Hunter-Gatherer Children Learn Social and Gender Norms? A Meta-Ethnographic Review. *Cross-Cultural Research*, *52*(2), 213–255. https://doi.org/10.1177/1069397117723552

Lew-Levy, S., Milks, A., Lavi, N., Pope, S. M., & Friesem, D. E. (2020). Where innovations flourish: An ethnographic and archaeological overview of hunter–gatherer learning contexts. *Evolutionary Human Sciences*, *2*, e31. https://doi.org/10.1017/ehs.2020.35

Lew-Levy, S., Reckin, R., Lavi, N., Cristóbal-Azkarate, J., & Ellis-Davies, K. (2017). How Do Hunter-Gatherer Children Learn Subsistence Skills? *Human Nature*, *28*(4), 367–394. https://doi.org/10.1007/s12110-017-9302-2

Little, K. L. (1951). *The mende of sierra leone: A west african people in transition* (First Edition). Routledge & Kegan Paul, Limited.

Liu, C. (2024). Variation Matters: Expanding the Scope of Experimental Archaeology. *Advances in Archaeological Practice*, *12*(4), 375–389. https://doi.org/10.1017/aap.2024.30

Liu, C., & Stout, D. (2023). Inferring cultural reproduction from lithic data: A critical review. *Evolutionary Anthropology: Issues, News, and Reviews*, *32*(2), 83–99. https://doi.org/10.1002/evan.21964

Lloyd, P. (1953). Craft organization in yoruba towns. *Africa*, *23*(1), 3044. https://www.cambridge.org/core/journals/africa/article/craft-organization-in-yoruba-towns1/91C914583A17E012C0C11CA586C9

Mair, L. P. (1934). *An African people in the twentieth century*. George Routledge & Sons. http://archive.org/details/africanpeopleint0000mair

Malafouris, L. (2012). Prosthetic gestures: how the tool shapes the mind. *Behavioral and Brain Sciences*, *35*(4), 230–231. https://doi.org/10.1017/S0140525X11001919

Malafouris, L. (2021). How does thinking relate to tool making? *Adaptive Behavior*, *29*(2), 107–121. https://doi.org/10.1177/1059712320950539

Marwick, B. (2017). Computational Reproducibility in Archaeological Research: Basic Principles and a Case Study of Their Implementation. *Journal of Archaeological Method and Theory*, *24*(2), 424–450. https://doi.org/10.1007/s10816-015-9272-9

Marzke, M. W. (2013). Tool making, hand morphology and fossil hominins. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *368*(1630), 20120414. https://doi.org/10.1098/rstb.2012.0414

Maynard, A. E., Greenfield, P. M., Childs, C. P., & Weinstock, M. (2024). Social change, cultural evolution, weaving apprenticeship, and development: Informal education across three generations and 42 years in a maya community. *Applied Developmental Science*, *28*(1), 82–105. https://doi.org/10.1080/10888691.2022.2151445

McGuigan, N., Gladstone, D., & Cook, L. (2012). Is the Cultural Transmission of Irrelevant Tool Actions in Adult Humans (Homo Sapiens) Best Explained as the Result of an Evolved Conformist Bias? *PLOS ONE*, *7*(12), e50863. https://doi.org/10.1371/journal.pone.0050863

Mead, M. (1964). *Continuities in cultural evolution*. Yale University Press.

Nadel, S. F. (1947). *The Nuba: An Anthropological Study of the Hill Tribes in Kordofan*. Oxford University Press.

Naik, T. B. (1956). *The Bhils: A Study*. Bharatiya Adimjati Sevak Sangh. http://archive.org/details/dli.ernet.51866

Nash, J. C. (1970). *In the eyes of the ancestors: Belief and behavior in a maya community*. Yale University Press.

Opler, M. E. (1969). *Apache odyssey: A journey between two worlds*. Holt, Rinehart; Winston.

Osiurak, F., Lasserre, S., Arbanti, J., Brogniart, J., Bluet, A., Navarro, J., & Reynaud, E. (2021). Technical reasoning is important for cumulative technological culture. *Nature Human Behaviour*, *5*(12), 1643–1651. https://doi.org/10.1038/s41562-021-01159-9

Osiurak, F., & Reynaud, E. (2020). The elephant in the room: What matters cognitively in cumulative technological culture. *Behavioral and Brain Sciences*, *43*, e156. https://doi.org/10.1017/S0140525X19003236

Osterhaus, C., & Koerber, S. (2021). The Development of Advanced Theory of Mind in Middle Childhood: A Longitudinal Study From Age 5 to 10 Years. *Child Development*, *92*(5), 1872–1888. https://doi.org/10.1111/cdev.13627

Perequin, P., & Petrequin, A.-M. (2020). *Ecology of a tool: The ground stone axes of irian jaya*. Oxbow Books.

Pospisil, L. (1958). *Kapauku papuans and their law*. Yale University Press.

Richards, A. I. (1939). *Land, labour and diet in Northern Rhodesia: An economic study of the Bemba tribe*. Oxford University Press. http://archive.org/details/landlabourdietin0000rich

Riede, F., Lew-Levy, S., Johannsen, N. N., Lavi, N., & Andersen, M. M. (2023). Toys as Teachers: A Cross-Cultural Analysis of Object Use and Enskillment in Hunter–Gatherer Societies. *Journal of Archaeological Method and Theory*, *30*(1), 32–63. https://doi.org/10.1007/s10816-022-09593-3

Schniter, E., Kaplan, H. S., & Gurven, M. (2023). Cultural transmission vectors of essential knowledge and skills among tsimane forager-farmers. *Evolution and Human Behavior*, *44*(6), 530–540. https://doi.org/10.1016/j.evolhumbehav.2022.08.002

Scoditti, G. M. G. (1989). *Kitawa: A Linguistic and Aesthetic Analysis of Visual Art in Melanesia*. Walter de Gruyter.

Smith, M. W. (1940). *The Puyallup-Nisqually*. Columbia University Press.

Sterelny, K. (2021). Veiled agency? Children, innovation and the archaeological record. *Evolutionary Human Sciences*, *3*, e12. https://doi.org/10.1017/ehs.2021.9

Stout, D. (2002). Skill and cognition in stone tool production: An ethnographic case study from irian jaya. *Current Anthropology*, *43*(5), 693–722. https://doi.org/10.1086/342638

Stout, D. (2021). The cognitive science of technology. *Trends in Cognitive Sciences*, *25*(11), 964–977. https://doi.org/10.1016/j.tics.2021.07.005

Stout, D., & Chaminade, T. (2007). The evolutionary neuroscience of tool making. *Neuropsychologia*, *45*(5), 1091–1100. https://doi.org/10.1016/j.neuropsychologia.2006.09.014

Stout, D., & Hecht, E. (2024). Evolutionary Neuroarchaeology. In T. Wynn, K. A. Overmann, & F. L. Coolidge (Eds.), *The Oxford Handbook of Cognitive Archaeology* (pp. 197–222). Oxford University Press. https://doi.org/10.1093/oxfordhb/9780192895950.013.14

Stout, D., & Khreisheh, N. (2015). Skill Learning and Human Brain Evolution: An Experimental Approach. *Cambridge Archaeological Journal*, *25*(4), 867–875. https://doi.org/10.1017/S0959774315000359

Teit, J. A. (1956). Field notes on the tahltan and kaska indians: 1912-1915. *Anthropologica*, *3*, 39–171. https://www.jstor.org/stable/25604386

Theodoratus, D. (1971). *Identity crises: Changes in life style of the manchester band of pomo* *indians* [PhD thesis].

Tierney, N. J., & Cook, D. H. (2020). *Expanding tidy data principles to facilitate missing data exploration, visualization and assessment of imputations*. arXiv. https://doi.org/10.48550/arXiv.1809.02264

Turnbull, C. M. (1965). *Wayward servants: The two worlds of the african pygmies* (First Edition). The Natural History Press.

Wakano, J. Y., & Aoki, K. (2007). Do social learning and conformist bias coevolve? Henrich and boyd revisited. *Theoretical Population Biology*, *72*(4), 504–512. https://doi.org/10.1016/j.tpb.2007.04.003

Watson-Franke, M.-B., & Watson, L. C. (1986). Conflicting images of women and female enculturation among urban guajiro. *Journal of Latin American Lore*, *12*(2), 141159. https://ehrafworldcultures.yale.edu/cultures/sc13/documents/017

Weiner, A. B. (1976). *Women of value, men of renown: New perspectives in Trobriand exchange*. University of Texas Press. http://archive.org/details/womenofvaluemeno0000wein

Wenger, E. (1998). *Communities of practice: Learning, meaning, and identity*. Cambridge University Press.

Wengrow, D., & Graeber, D. (2015). Farewell to the 'childhood of man': ritual, seasonality, and the origins of inequality. *Journal of the Royal Anthropological Institute*, *21*(3), 597–619. https://doi.org/10.1111/1467-9655.12247

Wilbert, J. (1976). To become a maker of canoes: An essay in Warao enculturation. In J. Wilbert (Ed.), *Enculturation in Latin American: An Anthology* (Vol. 37, pp. 303–358). UCLA Latin American Center Publications, University of California. https://ehrafworldcultures.yale.edu/cultures/ss18/documents/021

Wilbert, J. (1977). Navigators of the winter sun. In E. Benson (Ed.), *Sea in the Precolumbian World: Conference at Dumbarton Oaks, October 26 & 27th, 1974* (pp. 16–46). Dumbarton Oaks Research Library; Collections, Trustees for Harvard University. https://ehrafworldcultures.yale.edu/cultures/ss18/documents/019

Wilkinson, R. J. (1920). *Life and customs. Part i. The incidents of malay life* (Second edition). Kelly & Walsh.

Wilson, J., Andrews, G., Hogan, C., Wang, S., & Shum, D. H. K. (2018). Executive function in middle childhood and the relationship with theory of mind. *Developmental Neuropsychology*, *43*(3), 163–182. https://doi.org/10.1080/87565641.2018.1440296